Algebra 4 — Using formulae · Change of subject

Calculator allowed for the numerical parts. Marks in brackets. ★ = harder question.

1 Make x the subject of each equation. [6]

a $d - 4x = f$ **b** $p = 3(q + x)$ **c** $\dfrac{x + a}{b} = c$

2 Make x the subject of each formula. [9]

a $x^2 - B = 3C$ **b** $\dfrac{x^2}{y} - z = a$ **c** $p(x - q) = x$

3 The formula $s = ut + \frac{1}{2}at^2$ connects distance s, initial speed u, time t and acceleration a. [4]

a Make a the subject.

b Calculate a when $s = 120$ m, $u = 3$ m/s and $t = 6$ s.

4 A shape is a semicircle of radius r. [6]

a Write a formula for its perimeter P (curved edge plus diameter).

b Write a formula for its area A.

c Find the value of r for which $A = P$ numerically.

5 The volume of a cylinder is $V = \pi r^2 h$. A cylinder has volume 500 cm³ and height 10 cm. Find its radius. [3]

6 Make t the subject of $m = \dfrac{3t + 1}{t - 2}$. [4]

Total: 32 marks

Check: undo operations in reverse order, one step at a time · a square root gives ± unless the quantity must be positive · if the new subject appears twice, collect its terms and factorise it out.